

INSTITUT UNIVERSITAIRE DES SCIENCES

TD-SYSTEME#6

INNOCENT COFFY MANAHENN

L3

FST

```
94% [31 Packages 1 207 kB/1 510 kB 80%] 264 kB/s 3s5s
Réception de :32 http://archive.ubuntu.com/ubuntu noble-updates/universe Transl
96% [32 Translation-en 119 kB/307 kB 39%] 264 kB/s 2s
Réception de :33 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64
Réception de :34 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64
Réception de :35 http://archive.ubuntu.com/ubuntu noble-updates/multiverse amd6
Réception de :36 http://archive.ubuntu.com/ubuntu noble-backports/main amd64 Co
100% [36 Components-amd64 5 526 B/7 304 B 76%] 264 kB/s 0s
Réception de :37 http://archive.ubuntu.com/ubuntu noble-backports/restricted ar
Réception de :38 http://archive.ubuntu.com/ubuntu noble-backports/universe amd6
Réception de :39 http://archive.ubuntu.com/ubuntu noble-backports/multiverse ar
16,0 Mo réceptionnés en 36s (445 ko/s)
```

-v

```
Lecture des listes de paquets... Fait
Construction de l'arbre des dépendances... Fait
Lecture des informations d'état... Fait
39 paquets peuvent être mis à jour. Exécutez « apt list --upgradable » pour les
coffy@KRYXCVII:~$ sudo apt install bind9 bind9utils bind9-doc
Lecture des listes de paquets... Fait
Construction de l'arbre des dépendances... Fait
Lecture des informations d'état... Fait
bind9 est déjà la version la plus récente (1:9.18.39-0ubuntu0.24.04.2).
bind9utils est déjà la version la plus récente (1:9.18.39-0ubuntu0.24.04.2).
bind9-doc est déjà la version la plus récente (1:9.18.39-0ubuntu0.24.04.2).
0 mis à jour, 0 nouvellement installés, 0 à enlever et 39 non mis à jour.
coffy@KRYXCVII:~$ named -v
BIND 9.18.39-0ubuntu0.24.04.2-Ubuntu (Extended Support Version) <id:>
```

GNU nano 7.2

/etc/bind/named.conf.local

```
//  
// Do any local configuration here  
//  
  
zone "coffy.local" {  
    type master;  
    file "/etc/bind/db.coffy.local";  
};  
  
// Consider adding the 1918 zones here, if they are not used in your  
// organization  
//include "/etc/bind/zones.rfc1918";  
zone "1.168.192.in-addr.arpa" {  
    type master;  
    file "/etc/bind/db.192";  
};
```

Nom du fichier à écrire: /etc/bind/named.conf.local

^G Aide

M-D Format DOS

M-A Ajout (à la

```
coffy@KRYXCVII: ~  
GNU nano 7.2 /etc/bind/named.conf.local  
//  
// Do any local configuration here  
//  
  
zone "coffy.local" {  
type master;  
file "/etc/bind/db.coffy.local";  
};  
  
// Consider adding the 1918 zones here, if they are not used in your  
// organization  
//include "/etc/bind/zones.rfc1918";  
zone "1.168.192.in-addr.arpa" {  
type master;  
file "/etc/bind/db.192";  
};  
  
coffy@KRYXCVII: ~  
GNU nano 7.2 /etc/bind/named.conf.local *  
//  
// Do any local configuration here  
//  
  
zone "coffy.local" {  
type master;  
file "/etc/bind/db.coffy.local";  
};  
  
// Consider adding the 1918 zones here, if they are not used in your  
// organization  
//include "/etc/bind/zones.rfc1918";  
zone "1.168.192.in-addr.arpa" {  
type master;  
file "/etc/bind/db.192.168.1";  
};
```

```

: commande introuvable
KRYXCVII:~$ named -v
9.18.39-0ubuntu0.24.04.2-Ubuntu (Extended Support Version) <id:>
KRYXCVII:~$ sudo systemctl enable
Mot de passe de coffy :
v arguments.
KRYXCVII:~$ sudo systemctl enable
v arguments.
KRYXCVII:~$ sudo systemctl enable bind9
to enable unit: Refusing to operate on alias name or linked unit file: bind9.service
KRYXCVII:~$ sudo systemctl enable bind9.service
to enable unit: Refusing to operate on alias name or linked unit file: bind9.service
KRYXCVII:~$ sudo nano /etc/bind/named.conf.local
KRYXCVII:~$ sudo cp /etc/bind/db.local /etc/bind/db.coffy.local
KRYXCVII:~$ sudo nano /etc/bind/db.coffy.local
KRYXCVII:~$ sudo nano /etc/bind/db.coffy.local
KRYXCVII:~$ sudo nano /etc/bind/named.conf.local
KRYXCVII:~$ sudo cp /etc/bind/db.127 /etc/bind/db.192.168.1
KRYXCVII:~$ sudo nano /etc/bind/db.192.168.1
KRYXCVII:~$ sudo named-checkconf
/etc/bind/named.conf.options:15: missing ';' before '8.8.4.4'
/etc/bind/named.conf.options:15: unknown option '8.8.4.4'
KRYXCVII:~$ █

```

```

bind9 : commande introuvable
coffy@KRYXCVII:~$ named -v
BIND 9.18.39-0ubuntu0.24.04.2-Ubuntu (Extended Support Version) <id:>
coffy@KRYXCVII:~$ sudo systemctl enable
[sudo] Mot de passe de coffy :
Too few arguments.
coffy@KRYXCVII:~$ sudo systemctl enable
Too few arguments.
coffy@KRYXCVII:~$ sudo systemctl enable bind9
Failed to enable unit: Refusing to operate on alias name or linked unit file: bind9.service
coffy@KRYXCVII:~$ sudo systemctl enable bind9.service
Failed to enable unit: Refusing to operate on alias name or linked unit file: bind9.service
coffy@KRYXCVII:~$ sudo nano /etc/bind/named.conf.local
coffy@KRYXCVII:~$ sudo cp /etc/bind/db.local /etc/bind/db.coffy.local
coffy@KRYXCVII:~$ sudo nano /etc/bind/db.coffy.local
coffy@KRYXCVII:~$ sudo nano /etc/bind/db.coffy.local
coffy@KRYXCVII:~$ sudo nano /etc/bind/named.conf.local
coffy@KRYXCVII:~$ sudo cp /etc/bind/db.127 /etc/bind/db.192.168.1
coffy@KRYXCVII:~$ sudo nano /etc/bind/db.192.168.1
coffy@KRYXCVII:~$ sudo named-checkconf
/etc/bind/named.conf.options:15: missing ';' before '8.8.4.4'
/etc/bind/named.conf.options:15: unknown option '8.8.4.4'
coffy@KRYXCVII:~$ sudo named-checkzone coffy.local /etc/bind/db.coffy.local
zone coffy.local/IN: loaded serial 1
OK
coffy@KRYXCVII:~$ sudo named-checkzone 1.168.192.in-addr.arpa /etc/bind/db.192.168.1
zone 1.168.192.in-addr.arpa/IN: loading from master file /etc/bind/db.192.168.1 failed: file not found
zone 1.168.192.in-addr.arpa/IN: not loaded due to errors.
coffy@KRYXCVII:~$ █

```

```

coffy@KRYXCVII:~$ sudo named-checkconf
/etc/bind/named.conf.options:15: missing ';' before '8.8.4.4'
/etc/bind/named.conf.options:15: unknown option '8.8.4.4'
coffy@KRYXCVII:~$ sudo named-checkzone coffy.local /etc/bind/db.coffy.local
zone coffy.local/IN: loaded serial 1
OK
coffy@KRYXCVII:~$ sudo named-checkzone 1.168.192.in-addr.arpa /etc/bind/db.192.168.1
zone 1.168.192.in-addr.arpa/IN: loading from master file /etc/bind/db.192.168.1 failed: file not found
zone 1.168.192.in-addr.arpa/IN: not loaded due to errors.
coffy@KRYXCVII:~$ dig coffy.local

; <<>> DiG 9.18.39-0ubuntu0.24.04.2-Ubuntu <<>> coffy.local
;; global options: +cmd
;; Got answer:
;; WARNING: .local is reserved for Multicast DNS
;; You are currently testing what happens when an mDNS query is leaked to DNS
;; ->HEADER<- opcode: QUERY, status: SERVFAIL, id: 45579
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 0, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 65494
;; QUESTION SECTION:
;coffy.local.                IN      A

;; Query time: 0 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Thu Jan 08 16:44:20 EST 2026
;; MSG SIZE rcvd: 40

; EDNS: version: 0, flags:; udp: 65494
;; QUESTION SECTION:
;coffy.local.                IN      A

;; Query time: 0 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Thu Jan 08 16:44:20 EST 2026
;; MSG SIZE rcvd: 40

coffy@KRYXCVII:~$ dig pc1.coffy.local

; <<>> DiG 9.18.39-0ubuntu0.24.04.2-Ubuntu <<>> pc1.coffy.local
;; global options: +cmd
;; Got answer:
;; WARNING: .local is reserved for Multicast DNS
;; You are currently testing what happens when an mDNS query is leaked to DNS
;; ->HEADER<- opcode: QUERY, status: SERVFAIL, id: 50928
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 0, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 65494
;; QUESTION SECTION:
;pc1.coffy.local.           IN      A

;; Query time: 1 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Thu Jan 08 16:44:59 EST 2026
;; MSG SIZE rcvd: 44

```

```

; EDNS: version: 0, flags:; udp: 65494
;; QUESTION SECTION:
;pc1.coffy.local.                IN      A

;; Query time: 1 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Thu Jan 08 16:44:59 EST 2026
;; MSG SIZE rcvd: 44

coffy@KRYXCVII:~$ dig www.coffy.local

; <<>> DiG 9.18.39-0ubuntu0.24.04.2-Ubuntu <<>> www.coffy.local
;; global options: +cmd
;; Got answer:
;; WARNING: .local is reserved for Multicast DNS
;; You are currently testing what happens when an mDNS query is leaked to DNS
;; ->>HEADER<<- opcode: QUERY, status: SERVFAIL, id: 56297
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 0, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 65494
;; QUESTION SECTION:
;www.coffy.local.                IN      A

;; Query time: 1 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Thu Jan 08 16:45:25 EST 2026
;; MSG SIZE rcvd: 44

```

```

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 65494
;; QUESTION SECTION:
;www.coffy.local.                IN      A

;; Query time: 1 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Thu Jan 08 16:45:25 EST 2026
;; MSG SIZE rcvd: 44

coffy@KRYXCVII:~$ dig -x 192.168.1.20

; <<>> DiG 9.18.39-0ubuntu0.24.04.2-Ubuntu <<>> -x 192.168.1.20
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NXDOMAIN, id: 38732
;; flags: qr rd ra; QUERY: 1, ANSWER: 0, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 65494
;; QUESTION SECTION:
;20.1.168.192.in-addr.arpa.      IN      PTR

;; Query time: 172 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Thu Jan 08 16:45:57 EST 2026
;; MSG SIZE rcvd: 54

```

```

default-lease-time 600;
max-lease-time 7200;

authoritative;

subnet 192.168.10.0 netmask 255.255.255.0 {
    range 192.168.10.50 192.168.10.150;
    option routers 192.168.10.1;
    option subnet-mask 255.255.255.0;
    option domain-name "coffy.local";
    option domain-name-servers 192.168.1.100, 8.8.8.8;
}

host pc-fixe {
    hardware ethernet AA:BB:CC:DD:EE:FF;
    fixed-address 192.168.10.20;
}

```

[Lecture de 17 lignes]

^G Aide ^O Écrire ^W Chercher ^K Couper ^T Exécuter ^C Emplacement M-U Annuler

```

coffy@KRYXCVII:~$ sudo systemctl restart isc-dhcp-server
coffy@KRYXCVII:~$ sudo systemctl status isc-dhcp-server
* isc-dhcp-server.service - ISC DHCP IPv4 server
   Loaded: loaded (/usr/lib/systemd/system/isc-dhcp-server.service; enabled; preset: enabled)
   Active: failed (Result: exit-code) since Thu 2026-01-08 17:07:18 EST; 2s ago
     Duration: 283ms

```

```

janv. 08 17:07:18 KRYXCVII dnsmasq[12042]:
janv. 08 17:07:18 KRYXCVII dhcpcd[12042]: exiting.
janv. 08 17:07:18 KRYXCVII systemd[1]: isc-dhcp-server.service: Main process exited, code=exited, status=1/FAILURE
janv. 08 17:07:18 KRYXCVII systemd[1]: isc-dhcp-server.service: Failed with result 'exit-code'.

```


1-Le role des message dhcp

- **DHCPDISCOVER** : le client cherche un serveur DHCP.
- **DHCPOFFER** : le serveur propose une adresse IP.
- **DHCPREQUEST** : le client accepte l'offre.
- **DHCPACK** : le serveur confirme l'attribution.

2-Si deux serveur dhcp répondent en même temps

Le client reçoit plusieurs offres et choisit la première reçue. Cela peut provoquer des conflits d'adresses IP si le réseau n'est pas contrôlé.

3- quel est l'intérêt des réservations d'IP fixes

Garantir qu'un équipement reçoit toujours la même adresse IP, utile pour les serveurs, imprimantes et équipements réseau.

4-un reseaux pour 300 machines

Utiliser un réseau /23

Exemple :

- Réseau : 192.168.20.0/23
- Masque : 255.255.255.0
- Nombre d'hôtes : 510

Ce TD m'a permis de comprendre le fonctionnement des serveurs DNS et DHCP sous Linux. J'ai appris à créer des zones DNS, gérer la résolution directe et inverse, configurer un serveur DHCP et analyser les logs réseau.

Malgré quelques difficultés de configuration, les tests ont permis de valider le bon fonctionnement des services.